

## APPLICATION-BASED QUESTIONS – COLLEGE ASSIGNMENT

**Subject:** Cryptography and Network Security

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### Code of Conduct:

I Mohammed Arshad R/192521126 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**



## 1. Event Communication Security

[illegible]

cryptanalysis. These concepts are fundamental in protecting sensitive information over computer networks. Modern cryptography focuses on confidentiality, integrity, authentication, non-repudiation, secure key management, and resilience against cryptanalysis. These concepts are fundamental in protecting sensitive information over computer networks.

## 2. Intercepted Suspicious Message

[illegible]

### 3. Secure Messaging System

Using the Vigenère cipher with keyword **KEY**, HELLO becomes **RIJVS**. The keyword changes the shift for every character, providing greater security than Caesar cipher. However, repeated keys can still be attacked using statistical techniques. Modern cryptography focuses on confidentiality, integrity, authentication, non-repudiation, secure key management, and resilience against cryptanalysis. These concepts are fundamental in protecting sensitive information over computer networks. Modern cryptography focuses on







[illegible]